UNIT 2: LASER AND APPLICATIONS

LECTURE 2

Revision Lecture 1

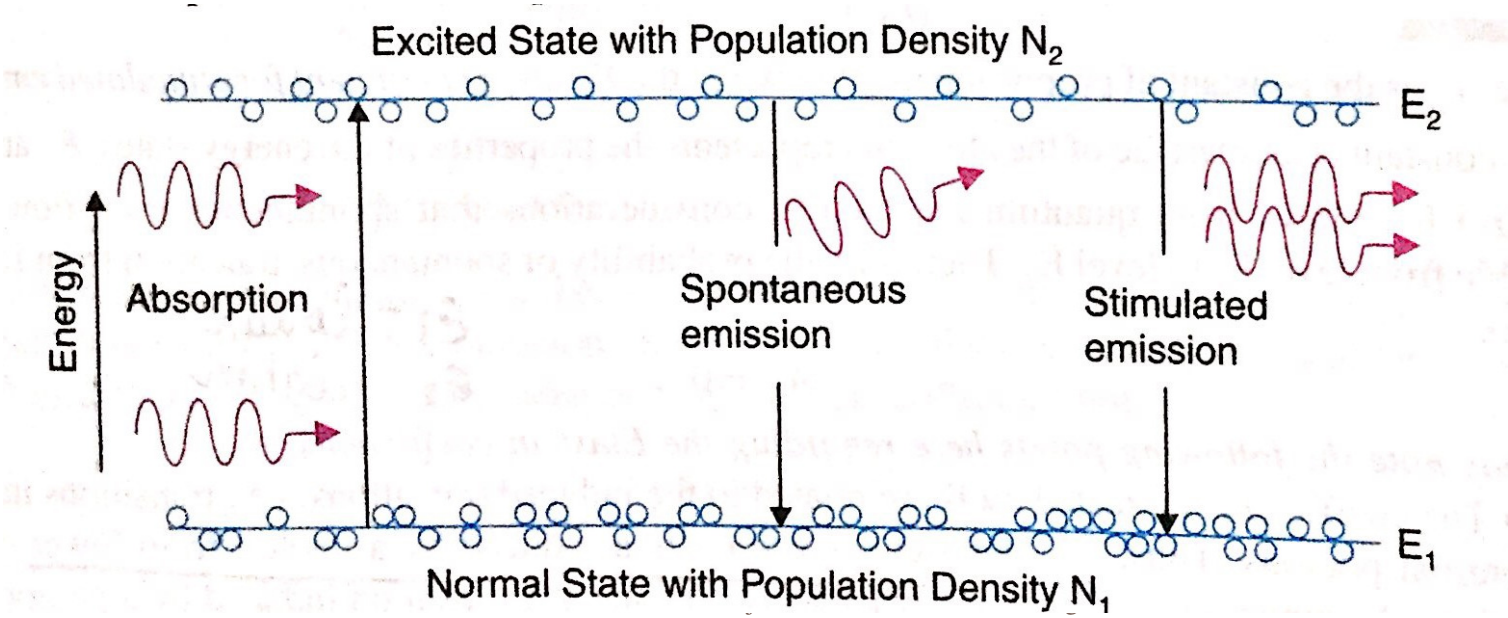
When photons of energy (hν) travel through a material three different processes occurs

1. **Absorption,**
2. **Spontaneous emission**
3. **Stimulated emission**

$$R_{abs} = P_{12}N_1 = B_{12}\,\rho(\nu)N_1$$

$$R_{sp} = P_{21}N_2 = A_{21}N_2$$

$$R_{st} = (P_{21})_{st}N_2 = B_{21}\,\rho(\nu)N_2$$



QUICK Quiz

Quick Quiz Response on the 8/29/2018 Lecture

No	Question	Attempts	Right	Wrong
1	LASER is the acronym of	69	38	31
2	DVD uses laser. True or false?	69	63	6
3	Which of the following is used in atomic clocks?	69	35	34

LASER is the acronym of

a) Light amplification and stimulated emission of radiation
b) Light amplification by stimulated emission of radiation
c) Light absorption by stimulated emission of radiation
d) Light absorption by spontaneous emission of radiation

Ans: B

Which of the following is used in atomic clocks?

a) Laser
b) Quartz
c) Maser
d) Helium

Ans: C

DVD uses laser. True or false?

a) True
b) False

Ans: A

LECTURE SCHEDULE

29/8/18 Lecture 1 ***Fundamentals of laser-*** energy levels in atoms, Radiation matter interaction, Absorption of light, Spontaneous emission of light, stimulated emission of light, Einstein A and B coefficients

30/8/18 Lecture 2: ***Fundamentals of Laser-*** Population of energy levels, Einstein's relation, Metastable state, population inversion, Resonant cavity

31/8/18 Lecture 3*: Fundamentals of Laser: Excitation mechanisms, lasing action, properties of laser*

5/9/18 Lecture 4:Type of Lasers**:** Nd - YAG, He-Ne Laser, Semiconductor Laser

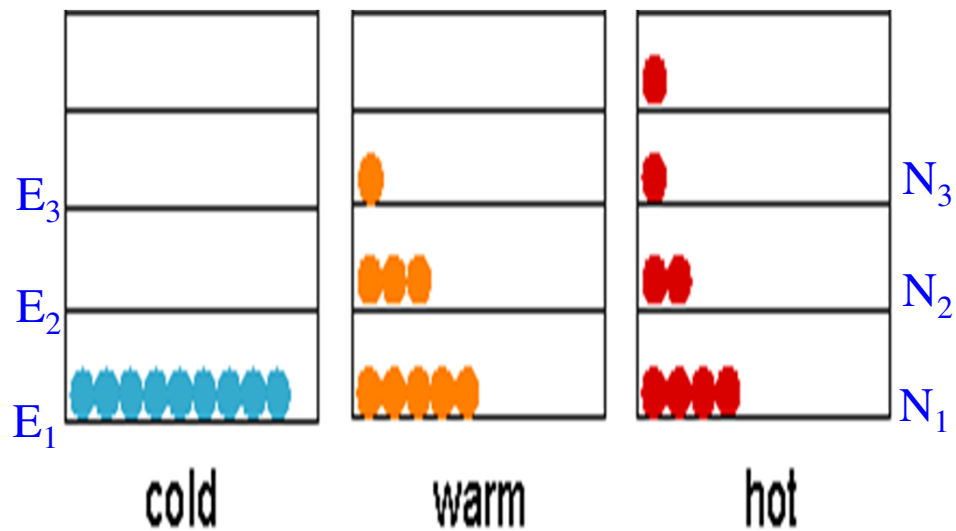
6/9/18 Lecture 5:,***Applications of laser*** in engineering, Holography.

7/9/18 Lecture 6:

POPULATION OF ENERGY LEVELS

Populations of energy levels is nothing but the total number of atoms occupying the particular Energy level (E).

$$\frac{N_1}{N_2} = \frac{e^{-E_1/kT}}{e^{-E_2/kT}} = e^{(E_2-E_1)/kT}$$



the *Boltzmann distribution* tells us that the ratio of populations varies exponentially with the energy difference, and the greater the level difference the smaller the population in the E_2 level.

EINSTEIN RELATIONS

Under steady state condition, Rate of absorption transitions and rate of emissions (induced as well as spontaneous) will balance each other

$$R_{abs} = R_{sp} + R_{st}$$

$$B_{12}\,\rho(\nu)N_1 = A_{21}\,N_2 + B_{21}\,\rho(\nu)N_2$$

Where, B_{12}, A_{21}, B_{21} are Einstein's coefficients for induced absorption, spontaneous emission and stimulated /induced emission respectively. N_1 and N_2 populations of atoms in the ground (E_1) are excited (E_2) states, respectively.

EINSTEIN RELATIONS

Just now we saw $B_{12}\,\rho(\nu)N_1 = A_{21}\,N_2 + B_{21}\,\rho(\nu)N_2$ under thermal equilibrium → Eq.1

Number of atoms absorbing photons per second per volume = Number of atoms emitting photons per second per volume

Re-arranging equation 1 we get,

$$\rho(\nu)[B_{12}\,N_1 - B_{21}N_2] = A_{21}\,N_2$$

So the photon density can be expressed as

$$\rho(\nu) = \frac{A_{21}\,N_2}{[B_{12}\,N_1 - B_{21}N_2]}$$ → Eq. 2

EINSTEIN RELATIONS

Now divide numerator and denominator by $B_{12}N_{2,}$ we get

$$\rho(\nu) = \frac{A_{21}/B_{12}}{\left[\frac{N_1}{N_2} - \frac{B_{21}}{B_{12}}\right]} \quad \text{----------} \rightarrow \quad \text{Eq. 3}$$

But Boltzmann's law, the distribution of atoms among the energy levels E_1 and E_2 at thermal equilibrium at Temperature T is

$$\frac{N_1}{N_2} = \frac{e^{-E_1/kT}}{e^{-E_2/kT}} = e^{(E_2-E_1)/kT} \qquad \text{And} \qquad E_2 - E_1 = h\nu$$

k Boltzmann's constant, h Planck's const and ν frequency of photon

Eqn.3 becomes

$$\rho(\nu) = \frac{A_{21}}{B_{12}} \frac{1}{[e^{h\nu/kT} - B_{21}/B_{12}]} \quad \text{----------} \rightarrow \quad \text{Eq. 4}$$

$$\rho(\nu) = \frac{A_{21}}{B_{12}} \frac{1}{[e^{h\nu/kT} - B_{21}/B_{12}]}$$ ----------→ Eq. 4

But according to Planck's law, the energy density of radiation $\rho(\nu)$, is given by the formula

$$\rho(\nu) = \left[\frac{8\pi h\nu^3\mu^3}{c^3}\right]\left[\frac{1}{[e^{h\nu/kT} - 1]}\right]$$ ----------→ Eq. 5

Wher μ is the refractive index of the medium and c is the velocity of light in free space. Photon energy density given by Eq.4 must be consistent with and Eq.5, then we get two relations for 3 Einstein's coefficients

$$\frac{A_{21}}{B_{12}} = \frac{8\pi h\nu^3\mu^3}{c^3}$$ ----------→ Eq. 6

$$\frac{B_{21}}{B_{12}} = 1 \qquad B_{21} = B_{12}$$ ----------→ Eq7

EINSTEIN RELATIONS

Eq.6 and **7** are known as **Einstein relation** and B_{12},B_{21} and A_{21} are **Einstein coefficients**

$$B_{12} = B_{21} = \frac{c^3}{8\pi h \nu^3 \mu^3} A_{21} \quad \text{----------} \rightarrow \text{Eq. 8}$$

Eq.8 Implied:

- probability of induced absorption and induced/stimulated emissions are same.. Means?
- Laser actions at high frequencies are difficult to achieve(x-rays onwards).. Why?

LASER

Light **A**mplification through (by) **S**timulated **E**mission of **R**adiation

So to realize LASER we need light amplification and for that stimulated emission is essential

But Under normal conditions

1. Induced absorption dominate stimulated emission for given a photon density. $N_1 >> N_2$
2. Spontaneous emission dominate stimulated emission for lower life time at the excited level. $\tau < 10^{-7}$s

How to get rid of these two issues to succeed in stimulated emission and hence light amplification?

Stimulated emission/spontaneous emission ($R_1=R_{st}/R_{sp}$)?

To get an idea about it let us find the typical value of $\mathbf{R_{st}/R_{sp}}$

From our last lecture we know, rate of stimulated transition (R_{st}) and rate of spontaneous transitions (R_{sp})are

$$R_{st} = B_{21}\,\rho(\nu)N_2 \qquad\qquad R_{sp} = A_{21}N_2$$

Now take the ratio of R_{st} to R_{sp}

$$R_1 = \frac{R_{st}}{R_{sp}} = \frac{B_{21}\,\rho(\nu)N_2}{A_{21}N_2} = \frac{B_{21}}{A_{21}}\rho(\nu) \quad \text{----------\!\!\!

Stimulated emission/ spontaneous emission ($R_1=R_{st}/R_{sp}$)?

Now Eq.9 changes to

$$R_1 = \frac{B_{21}}{A_{21}}\left[\frac{8\pi h\nu^3\mu^3}{c^3}\right]\left[\frac{1}{[e^{h\nu/kT}-1]}\right] \quad \text{----------\!\!\!

How to increase $R_1=R_{st}/R_{sp}$ to realize LASER?

We have $R_1 = \frac{B_{21}}{A_{21}}\rho(\nu)$; so when the photon density (ρ) and ratio of Einstein coefficient (B_{21}/A_{21}) are large, stimulated emission will dominate

1. But as the radiation density (ρ) increase absorption also increases, due to $B_{21}=B_{12}$. Hence large photon density of course help but not enough for more stimulated emissions…. ***Optical cavity?***
2. If the excited state has more life time ($1/A_{21}$ represents the lifetime of the excited state), R_1 increases many fold and stimulated emission increases substantially…. ***Metastable state***?

So an increase in the photon density (ρ) and the life time of atoms in the excited state, along with the dominance over **absorption transition** may work out for increasing the stimulated transition.

How to increase $R_2=R_{st}/R_{abs}$ to realize LASER?

$$R_2 = \frac{R_{st}}{R_{abs}} = \frac{B_{21}\,\rho(\nu)N_2}{B_{12}\,\rho(\nu)N_1}$$

$R_{21}=R_{12}$

$$R_2 = \frac{N_2}{N_1}$$ ----------→ **Eq. 12**

i.e. stimulated transition overcome absorption transition if $N_2 >> N_1$.

3. Unfortunately, under normal condition population in the excited level (N_2) is very much lower than that in the ground state(N_1). We have to invert this situation…. ***Population inversion?***

So we need three different requirements for getting dominant stimulated transition and LASER action

- **(i) Large photon density (ρ) –Optical resonant cavity**
- **(ii) Large life time of atoms in the excited state- Metastable state**
- **(iii) Large number of excited atoms - population inversion**

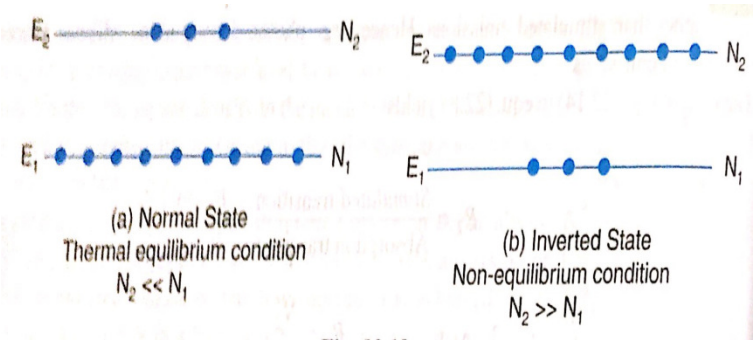
How to attain these three requirements? we will see next….

Meeting the requirements?

1. POPULATION INVERSION by Pumping

When the material is at thermal equilibrium, population ratio between E_1 and E_2 is given by the Boltzmann's law

$$\frac{N_2}{N_1} = e^{-(E_2-E_1)/kT}$$

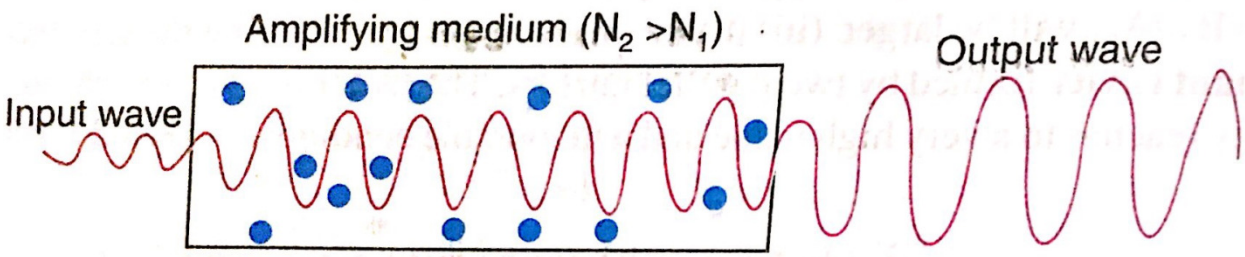


Negative temperature to achieve ($N_2 >> N_1$) population inversion? Lowest temperature one can achieve is T=0° K(-273°C)

So one has to supply energy from outside to attain $N_2 >> N_1$ and hence population inversion occurs…. *There are different* ***excitation mechanisms*** *which we will see later*

1. POPULATION INVERSION by Pumping

In the inverted case, stimulated transition is triggered and photon multiplication occurs and hence light amplification happens



Amplification of a light wave in a medium with population inversion.

However, due to continues stimulated emission, population of E_2 reduces and this action comes to an end. To sustain one has to continuously excite atoms from E_1 to E_2 and this process is called **pumping or optical pumping**– like you pump water to your tank for continuous usage of water☺

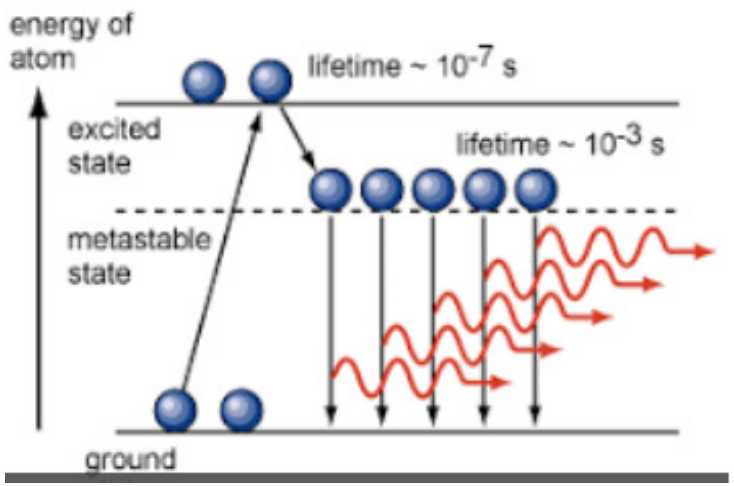
Meeting the requirements?

2. METASTABLE STATE to increase life time

After achieving population inversion, one has to suppress spontaneous emission that generally happens in nano seconds time

To increase the population at the excited state, life time at that state should be increased 10^{-6} to 10^{-3} s..

This is achieved by allowing the excited atoms at the pumping level to loose fraction of the energy and jump to another level where they stay longer time.. This third state is called **Metastable state**

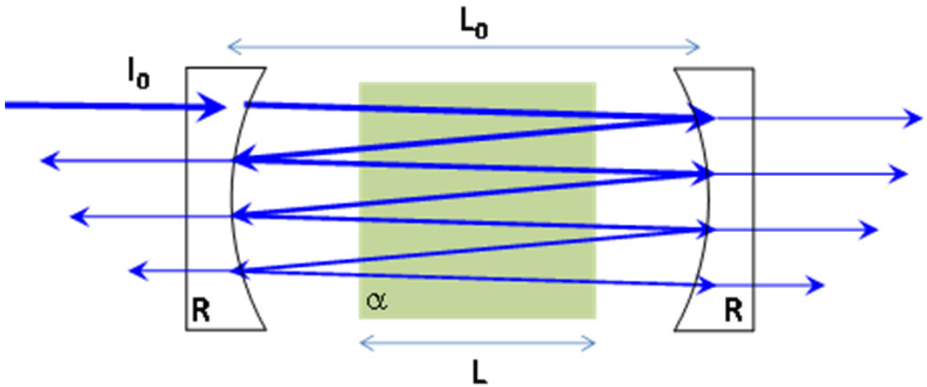


This is possible in material by doping

Meeting the requirements?

3. Large photon density (ρ) –Optical resonant cavity

To make stimulated emission to overtake spontaneous emission



That can be achieved

- By placing laser medium (material) between two mirrors
- Photon density build up due to repeated reflections
- Contained in the cavity to increase interaction- confinement

Quick Quiz

In normal conditions, absorption is more probable, and hence spontaneous emission dominate the stimulated emission.. Why?

a) More atoms are in the ground state compared to excited state.
b) Less atoms are in the excited state compared to ground state.
c) All of the above.
d) None of the above.

Einstein coefficient B_{12} for the absorption transition is equal to B_{21} of the stimulated transition .. Means?

a) When atoms are placed in a radiation field probability upward transition is more
b) When atoms are placed in a radiation field probability downward transition is more
c) Probability for downward and upward transition are equal
d) None of the above

Laser actions at high frequencies are difficult to achieve (x-rays onwards).. Why?

a) Coefficient of stimulated emission B_{21} is inversely proportional to the cube of the frequency
b) Coefficient of stimulated emission B_{21} is directly proportional to the cube of the frequency
c) Coefficient of stimulated emission B_{21} is nothing to do anything with frequency
d) None of the above

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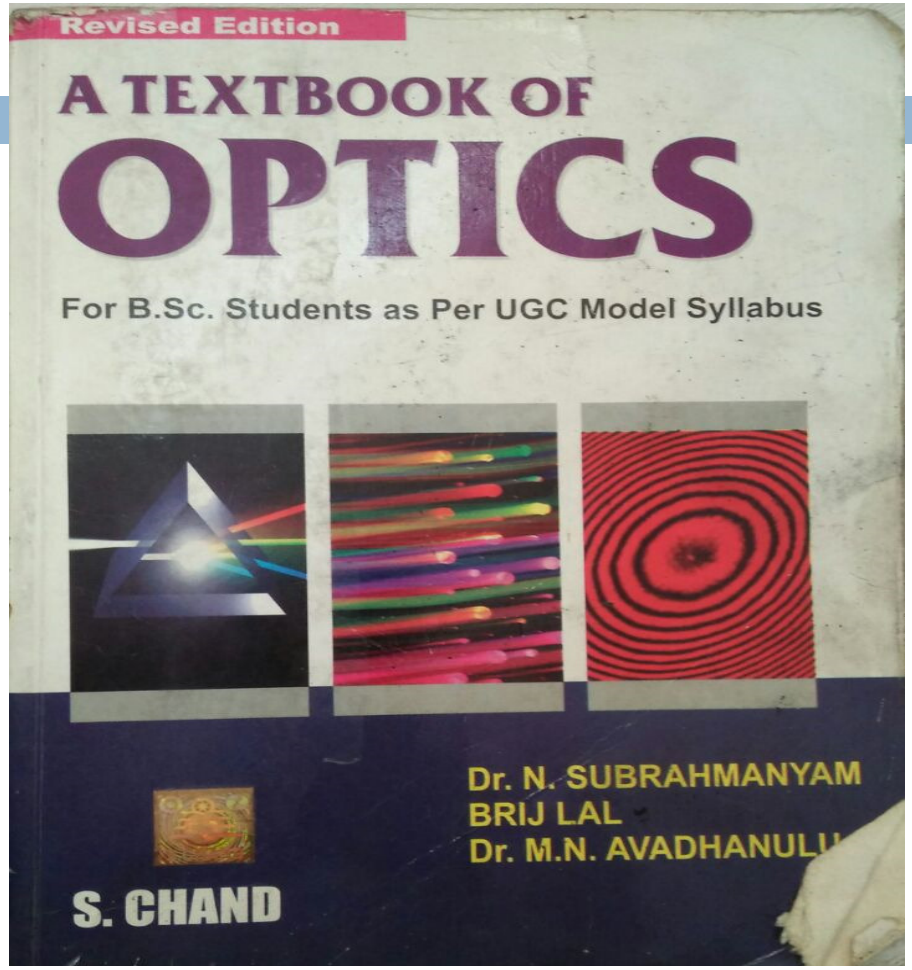
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